Kaan Aykurt

PhD Candidate · AI-driven Networked Systems · Technical University of Munich

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Research Profile

PhD candidate working at the intersection of **AI for Networking** and **Networking for AI**. I design agentic LLM workflows and benchmarking frameworks for autonomous networks, and study communication bottlenecks in large-scale AI/ML systems. My expertise includes reproducible benchmarking on hardware testbeds, transport layer analysis, and network modeling with Graph Neural Networks.

Experience

Chair of Communication Networks, Technical University of Munich 2022 – Now
Research and Teaching Assistant, Munich, Germany

- Developing agentic workflows for network troubleshooting, implementing the Model Context Protocol (MCP) for tool calling to interact with network emulators.
- Designed NetLLMBench, a reproducible benchmarking framework using Python and LangChain to evaluate LLM agents on network configuration tasks.
- Built and operated a Kubernetes experimental cluster and developed a custom pod autoscaler (HyPA) to optimize resource provisioning.
- Ranked 3rd in the GNNet2023 Challenge by designing Graph Neural Networks (GNNs) in PyTorch to predict latency on real-world datasets.
- Conducted hardware testbed measurements of TCP/IP and QUIC to analyze congestion dynamics in high-throughput applications.

Chair of Communication Networks, Technical University of Munich 2020 – 2022
Working Student, Munich, Germany

- Characterized traffic patterns of distributed training workloads (PyTorch DDP, MPI) to evaluate synchronization performance across RoCE and standard Ethernet.
- Modeled TCP congestion behavior in reconfigurable data centers to predict flow completion times and quantify topology switching impact.

IBM 2018 – 2019
IT Specialist, Istanbul, Türkiye

- Implemented automation workflows using Bash scripting and configured enterprise monitoring dashboards for operational insights.

Technical Skills

AI & ML: Python, PyTorch, Distributed Fine-Tuning, Hugging Face (Accelerate, PEFT/LoRA), vLLM
Networking: C++, RDMA (InfiniBand/RoCE), P4, eBPF, ns-3, Mininet, TCP/IP
Infrastructure: Docker, Kubernetes, Helm, Linux, Bash, Git, GitLab CI

Education

Technical University of Munich (TUM) 2022 – Now
PhD Computer Engineering (Advisor: Prof. Dr.-Ing. W. Kellerer)

Technical University of Munich (TUM) 2019 – 2022
M.Sc. Communications Engineering

Koç University 2014 – 2019
B.Sc. Electrical & Electronics Engineering | B.A. Economics

Selected Publications

[Full List on kaanaykurt.me]

- **RouteMind: An LLM-Based Framework for Inter-Domain Routing** *IEEE GLOBECOM, 2026*
- **NetLLMBench: A Benchmark Framework for LLMs in Networking** *IEEE NFV-SDN, 2024*
- **When TCP Meets Reconfigurations: A Measurement Study** *IEEE TNSM, 2023*
- **Network Traffic Characteristics of ML Frameworks** *IEEE CNSM, 2021*